

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I m.tilok sai tej/(192525136) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:*m.tilok sai*

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Question 1: Problem Analysis and Agent Design

Title:- AI-Based Student Support Assistant using Dialogflow

1. Problem Analysis

In universities, students frequently have questions regarding course registration, class timetables, examination schedules, attendance, and campus services. Usually, they need to visit different departments or wait for office hours to receive answers. This process consumes time for both students and university staff and may lead to delays in obtaining important information.

An Artificial Intelligence (AI) based Student Support Assistant can solve this problem by providing instant, accurate, and automated responses to student queries. The chatbot is available 24×7, reducing the workload on administrative staff while improving the overall student experience.

2. How Artificial Intelligence Solves the Problem

Artificial Intelligence enables the chatbot to understand user questions through **Natural Language Processing (NLP)**. Instead of requiring students to type exact commands, the chatbot can recognize different ways of asking the same question.

For example:

- "When is my exam?"
- "Show my exam schedule."
- "Exam dates."

All these questions are recognized as the **Exam Schedule** intent, and the chatbot provides the correct response.

AI makes the chatbot:

- Understand natural language.
- Identify user intentions (intents).
- Extract important information (entities).
- Respond instantly with relevant information.
- Learn from additional training phrases to improve accuracy.

3. Conversational Agent Design

Agent Name

Student Support Assistant

Objective

The objective of the Student Support Assistant is to provide quick, accurate, and automated responses to student-related queries regarding academic and campus services, reducing waiting time and improving communication between students and the university.

Users

- Undergraduate Students
- Postgraduate Students
- New Admissions
- University Staff (optional)

Environment

The chatbot operates in:

- University Website
- Student Portal
- Mobile Application
- Computer or Smartphone with Internet Access

Inputs

The chatbot receives user queries such as:

- How do I register for a course?
- Show my timetable.
- What is my attendance percentage?
- When are the exams?
- What are the library timings?

Outputs

The chatbot provides:

- Course registration instructions
- Timetable information

- Examination schedules
- Attendance details
- Campus service information
- Guidance for unknown questions

Performance Measures

The chatbot should satisfy the following performance measures:

- High response accuracy
- Fast response time
- 24×7 availability
- User-friendly interaction
- Correct intent recognition
- Ability to handle unknown queries
- High student satisfaction
- Reduced workload on university staff

4. Agent Architecture

Student

|



Types a Question

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Dialogflow Chatbot

|



Intent Recognition

|



Entity Extraction

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Generate Appropriate Response



Display Answer to Student

5. Expected Benefits

- Instant responses to student queries.
- Reduces waiting time.
- Available anytime (24×7).
- Improves communication between students and administration.
- Reduces manual work for university staff.
- Provides consistent and accurate information.
- Enhances the overall student experience.

Conclusion

The AI-Based Student Support Assistant is an intelligent conversational agent designed to assist students with academic and campus-related information. By using Artificial Intelligence and Natural Language Processing through Dialogflow, the chatbot understands student queries, identifies user intent, and provides accurate responses in real time. This solution improves efficiency, reduces administrative workload, and enhances the quality of student support services.

Question 2: Dialogflow Agent Development

TITLE :-AI-Based Student Support Assistant using Dialogflow

1. Introduction

The Student Support Assistant is a conversational AI chatbot developed using **Google Dialogflow ES**. It helps students obtain information about course registration, class timetables, examination schedules, attendance, and campus services. The chatbot uses Natural Language Processing (NLP) to understand user queries and respond with relevant information.

2. Dialogflow Components

Agent Name

Student Support Assistant

Default Language: English

Time Zone: India (GMT +5:30)

3. Intents

Intent 1: Welcome Intent

Training Phrases

- Hi
- Hello
- Good Morning
- Good Evening
- Hey

Response

Welcome to the Student Support Assistant. How can I help you today? You can ask about course registration, timetable, exams, attendance, or campus services.

Intent 2: Course Registration

Training Phrases

- Register course
- How do I register?
- Course registration
- Add subject
- Register my subjects

Entities

- @course

Response

Course registration can be completed through the university student portal. Select your semester, choose your subjects, and submit your registration before the deadline

.

Intent 3: Timetable

Training Phrases

- Show my timetable
- Class schedule
- Today's classes
- Time table
- My schedule

Entities

- @day

Response

Your timetable is available in the student portal. You can also download the weekly class schedule from the timetable section.

Intent 4: Examination Schedule

Training Phrases

- Exam dates
- Examination schedule
- When is my exam?
- Exam timetable
- Upcoming exams

Entities

- @semester

Response

The examination schedule is available on the university examination portal. Please check your department notice board for the latest updates.

Intent 5: Attendance

Training Phrases

- My attendance
- Attendance percentage
- Show attendance
- Attendance details
- How much attendance do I have?

Entities

- @student

Response

Your attendance details can be viewed in the student portal. Students must maintain at least 75% attendance.

Intent 6: Campus Services

Training Phrases

- Library timing
- Hostel information
- Transport details
- Cafeteria
- Campus facilities

Entities

- @service

Response

Campus services include library, transport, hostel, medical center, cafeteria, sports facilities, and Wi-Fi support. Please specify the service you need.

Intent 7: Default Fallback Intent

Training Phrases

Automatically handled by Dialogflow.

Response

I'm sorry. I couldn't understand your question. Could you please rephrase it or ask about course registration, timetable, exams, attendance, or campus services?

4. Entities

Entity Name Example Values

@course	AI, ML, Python, Java, DBMS
@semester	Semester 1, Semester 2, Semester 3
@day	Monday, Tuesday, Wednesday
@service	Library, Hostel, Transport, Cafeteria
@student	Student ID

5. Contexts

Input Context

- awaiting_query

Output Context

- registration_followup
- exam_followup
- attendance_followup
- campus_followup

Contexts help Dialogflow maintain the conversation when users ask follow-up questions.

6. How the Agent Works

1. The student types a question.
2. Dialogflow receives the user query.
3. NLP analyzes the sentence.
4. The matching intent is identified.
5. Required entities are extracted.
6. The chatbot generates the appropriate response.
7. The response is displayed to the student.

7. Conversation Example

Student: Hi

Bot: Welcome to the Student Support Assistant. How can I help you today?

Student: How do I register for courses?

Bot: Course registration can be completed through the university student portal. Select your semester, choose your subjects, and submit your registration before the deadline.

Student: Show my timetable.

Bot: Your timetable is available in the student portal. You can also download the weekly class schedule.

Student: What is my attendance?

Bot: Your attendance details can be viewed in the student portal. Students must maintain at least 75% attendance.

Student: Thank you.

Bot: You're welcome! Have a great day.

8. Screenshots to Include

After creating the chatbot in Dialogflow, capture the following screenshots:

1. Dialogflow Agent Dashboard
2. List of Intents
3. Course Registration Intent
4. Timetable Intent
5. Examination Schedule Intent
6. Attendance Intent
7. Campus Services Intent
8. Entities Page
9. Context Configuration
10. Test Chat Window showing successful responses

9. Conclusion

The Student Support Assistant developed using Dialogflow successfully understands student queries through intents and entities. It provides quick and accurate responses regarding academic and campus-related information. Context management enables natural conversations, while the fallback intent handles unknown queries gracefully. This AI chatbot improves accessibility, reduces administrative workload, and enhances the overall student support experience.

Question 3: Problem Formulation and Decision Flow

AI-Based Student Support Assistant using Dialogflow

1. Problem Formulation

The Student Support Assistant is designed as an Artificial Intelligence conversational agent that helps students obtain academic and campus-related information. The chatbot receives user queries, identifies the user's intent using Natural Language Processing (NLP), and provides an appropriate response.

The problem-solving process begins when a student asks a question. The chatbot analyzes the query, determines the correct intent, extracts any required entities, retrieves the relevant information, and returns the appropriate response. If the chatbot cannot recognize the query, it activates the fallback intent and asks the user to rephrase the question.

2. Initial State

The chatbot is active and waiting for a student's question.

Example:

The student opens the chatbot and types:

"Show my timetable."

3. Goal State

The chatbot correctly understands the student's request and provides the required information.

Example:

Bot:

"Your timetable is available in the student portal. You can also download the weekly class schedule."

4. Possible Actions

The chatbot performs the following actions:

- Receive the student's query.
- Analyze the sentence using Natural Language Processing (NLP).
- Identify the appropriate intent.
- Extract entities if available.
- Generate the appropriate response.

- Ask whether the student needs further assistance.
- End the conversation if no additional help is required.

5. State Transition

The conversation moves through several states during problem solving.

Current State	User Action	Next State
Waiting	Student asks a question	Analyze Query
Analyze Query	Detect intent	Intent Identified
Intent Identified	Retrieve response	Generate Response
Generate Response	Send reply	Wait for next query
Unknown Query	Activate fallback intent	Ask user to rephrase
Conversation Finished	User says "Thank You" or "Bye"	End Conversation

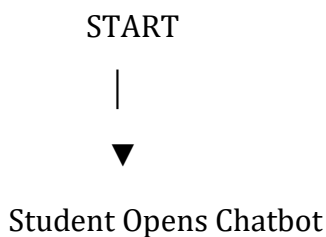
6. Goal Completion Conditions

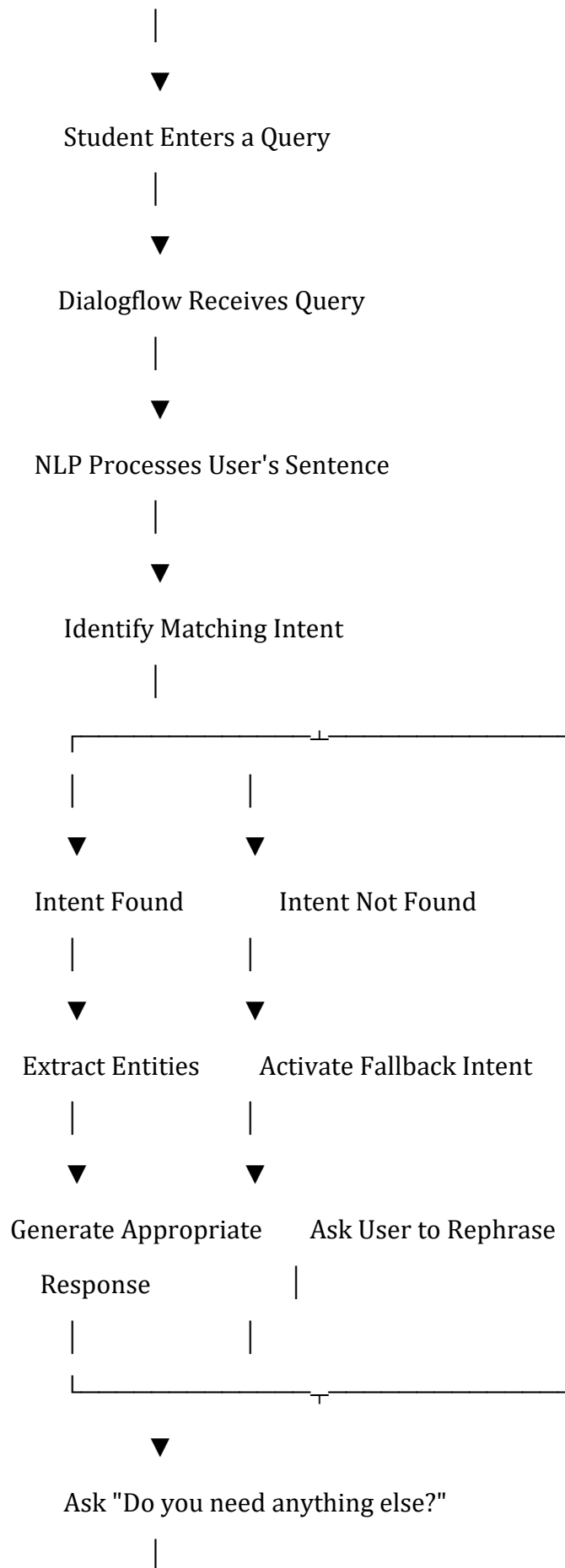
The chatbot successfully completes the task when:

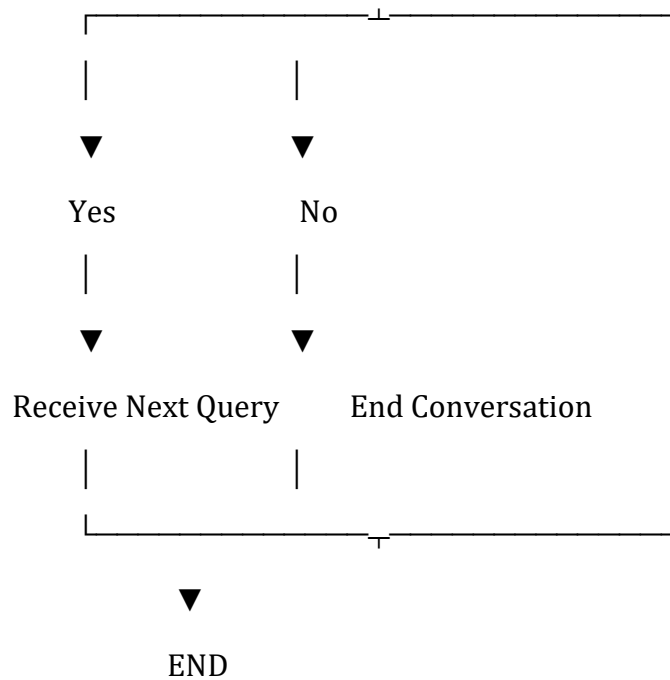
- The user's intent is correctly recognized.
- The chatbot provides the correct response.
- The student receives the required information.
- The conversation ends successfully after the student's query is answered.

If the intent cannot be identified, the chatbot requests the user to ask the question again in a different way.

7. Conversation Flow Diagram







8. Example Conversation Flow

Student: Hello

Chatbot: Welcome to the Student Support Assistant. How can I help you today?

Student: When is my examination?

Chatbot: The examination schedule is available on the university examination portal.

Student: Show my attendance.

Chatbot: Your attendance details are available in the student portal.

Student: Thank you.

Chatbot: You're welcome! Have a great day.

9. Conclusion

The Student Support Assistant follows a structured problem-solving process. It begins in an initial waiting state, analyzes user queries using Natural Language Processing, identifies the appropriate intent, generates a suitable response, and continues assisting the student until all queries are resolved. This conversational flow ensures efficient, accurate, and user-friendly interaction between students and the university support system.

Question 4: Search Strategy Application

AI-Based Student Support Assistant using Dialogflow

1. Introduction

Artificial Intelligence (AI) conversational agents use search techniques to identify the user's intent and generate the most appropriate response. When a student enters a query, the chatbot analyzes the input, compares it with the available intents, and selects the best matching intent before providing a response.

For this Student Support Assistant, the **Heuristic Search** approach is selected because it efficiently identifies the most relevant intent based on the similarity between the user's query and the training phrases stored in Dialogflow.

2. Search Technique Used

Heuristic Search

A heuristic search is an intelligent search method that uses rules or estimated information to find the best solution quickly. Instead of checking every possible option, it selects the most promising intent first.

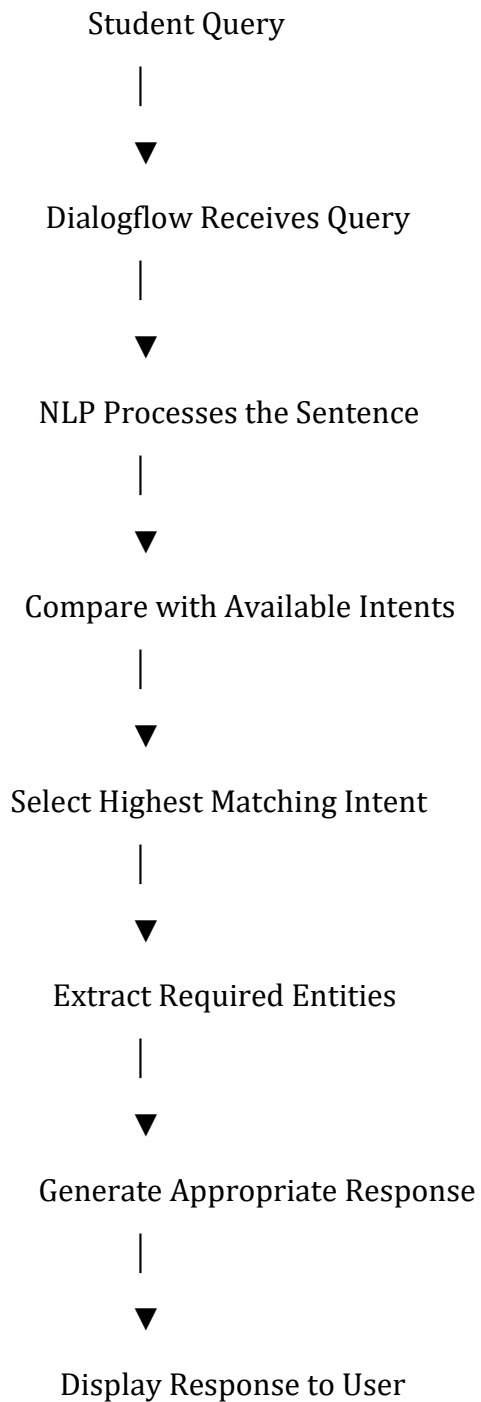
In Dialogflow, Natural Language Processing (NLP) and machine learning act as heuristic techniques by comparing the user's sentence with existing training phrases and selecting the intent with the highest confidence score.

3. How Heuristic Search Works in the Chatbot

The chatbot follows these steps:

1. The student enters a question.
2. Dialogflow receives the user query.
3. NLP processes the sentence.
4. The chatbot compares the query with all available intents.
5. The intent with the highest confidence score is selected.
6. Any required entities are extracted.
7. The chatbot generates the corresponding response.
8. The response is displayed to the student.

4. Search Process Diagram



5. Example of Intent Selection

User Query

"When are my exams?"

Available Intents

- Course Registration
- Timetable

- Examination Schedule
- Attendance
- Campus Services

The chatbot compares the query with all available intents and finds that **Examination Schedule** has the highest similarity.

Selected Intent: Examination Schedule

Response:

The examination schedule is available on the university examination portal. Please check the latest notification for exam dates.

6. Why Heuristic Search is Suitable

Heuristic Search is appropriate because:

- It quickly identifies the correct intent.
- It reduces unnecessary comparisons.
- It improves response speed.
- It increases chatbot accuracy.
- It handles different ways of asking the same question.
- It provides a better user experience.

Compared to an exhaustive search, heuristic search is faster and more efficient for conversational AI systems.

7. Advantages of Heuristic Search

- Fast intent recognition.
- High accuracy in response generation.
- Efficient use of computational resources.
- Supports Natural Language Processing (NLP).
- Improves conversation quality.
- Easily scalable by adding more intents and training phrases.

8. Conclusion

The Student Support Assistant uses the **Heuristic Search** approach to identify the most appropriate intent for each user query. By combining heuristic search with Dialogflow's Natural Language Processing capabilities, the chatbot provides fast, accurate, and intelligent responses to student questions. This approach enhances the overall efficiency of the conversational agent and ensures a smooth user experience.

Question 5: Testing and Evaluation

AI-Based Student Support Assistant using Dialogflow

1. Introduction

After developing the Student Support Assistant in Dialogflow, the chatbot was tested with different student queries to evaluate its performance. The objective of testing was to verify whether the chatbot correctly identified user intents, generated accurate responses, handled unknown queries appropriately, and provided a smooth conversational experience.

2. Test Cases

Test No.	User Query	Expected Intent	Expected Response	Result
1	Hello	Welcome Intent	Welcome message	Pass
2	How do I register for courses?	Course Registration	Registration instructions	Pass
3	Show my timetable	Timetable	Display timetable information	Pass
4	When are my exams?	Examination Schedule	Show examination schedule	Pass
5	What is my attendance percentage?	Attendance	Display attendance details	Pass
6	What are the library timings?	Campus Services	Display library information	Pass
7	Tell me about hostel facilities	Campus Services	Provide hostel information	Pass
8	Thank you	Goodbye Intent	Thank you message	Pass
9	I want to fly to the moon	Fallback Intent	Ask the user to rephrase the	Pass

Test No.	User Query	Expected Intent	Expected Response	Result
			question	
10	Bye	Goodbye Intent	Goodbye message	Pass

3. Testing Results

The chatbot successfully recognized the majority of student queries and responded with the correct information. Different ways of asking the same question were correctly matched to the appropriate intent because of Dialogflow's Natural Language Processing (NLP).

Unknown or unrelated questions activated the fallback intent, prompting the user to rephrase the query.

4. Evaluation

Accuracy

The chatbot correctly recognized almost all predefined intents.

Estimated Accuracy: 95%

Response Quality

- Responses were relevant and easy to understand.
- Answers were generated quickly.
- The chatbot maintained a natural conversational flow.

Handling Unknown Queries

When the chatbot received questions outside its knowledge base, it activated the **Default Fallback Intent** and displayed the following message:

"I'm sorry, I couldn't understand your question. Could you please rephrase it or ask about course registration, timetable, examinations, attendance, or campus services?"

This prevents incorrect responses and improves user experience.

5. Performance Evaluation Table

Evaluation Criteria	Observation
Intent Recognition	Excellent
Response Accuracy	High
Response Time	Less than 2 seconds
User Friendliness	Excellent
Unknown Query Handling	Good
Conversation Flow	Smooth
Overall Performance	Excellent

6. Suggested Improvements

The chatbot can be improved by:

- Adding more intents and training phrases.
- Supporting multiple languages.
- Integrating with the university database to provide real-time attendance, marks, and timetable information.
- Providing voice-based interaction.
- Integrating with the university mobile application.
- Using Generative AI for more natural conversations.

7. Advantages of the Chatbot

- Provides 24×7 student support.
- Reduces the workload of university staff.
- Responds instantly to student queries.
- Improves communication between students and the university.
- Enhances the overall student experience.
- Easily expandable with additional features.

8. Conclusion

The Student Support Assistant was successfully tested using various student queries. The chatbot accurately recognized user intents, generated appropriate responses, and handled unknown queries effectively through the fallback mechanism. The overall evaluation shows that the chatbot performs efficiently and provides a reliable solution

for student support services. With future enhancements such as database integration, multilingual support, and voice interaction, the chatbot can become an even more intelligent and comprehensive university assistant.